Google Summer of Code Proposal

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Idea Name: Physics Guided Machine Learning on Real Lensing Images

Learning the Lens: A Physics-Informed Framework for Dark Matter Substructure Classification

This project proposes a **physics-informed deep learning framework** for analyzing strong gravitational lensing images. The core idea is to integrate gravitational lensing equations directly into the learning pipeline through a differentiable simulator, enabling the model to learn representations that are both predictive and physically consistent.

The framework jointly performs image classification and lens parameter inference, producing interpretable outputs along with accurate predictions. Importantly, the design is **modular and architecture-agnostic**, allowing different encoder architectures to be used without modifying the physics-informed components.

The resulting system is designed for integration into the **DeepLense ecosystem**, enabling scalable and interpretable analysis of gravitational lensing datasets. While evaluation will focus on currently available data, the framework is designed to scale to future large survey data.

1 Abstract

Strong gravitational lensing provides a powerful probe for detecting dark matter substructures, but existing machine learning approaches often lack interpretability and do not incorporate known physical constraints.

This project introduces a physics-informed learning framework that embeds gravitational lensing equations into a differentiable simulation module. An encoder extracts latent representations from lensed images, enabling both classification and prediction of physical lens parameters.. These parameters are used to reconstruct the observed image via a differentiable lensing process.

By jointly optimizing classification and reconstruction objectives, the model learns representations that are both discriminative and physically meaningful. Preliminary results demonstrate that the framework achieves classification performance comparable to strong baselines while providing accurate parameter estimates and high-quality reconstructions.

The proposed approach establishes a flexible foundation for integrating physical modeling with deep learning and can be extended to more complex lens models and real observational data.

2 About Me

I am an undergraduate Computer Science student at the Indian Institute of Technology (IIT) Palakkad with strong interests in **scientific machine learning, physics-informed neural networks, and computational astrophysics**. My relevant academic coursework includes linear algebra, multivariable calculus, probability and statistics, optimization theory, special relativity.

My primary experience lies in developing **physics-informed deep learning systems**. For the DeepLense evaluation task, I designed and implemented a physics-informed neural network architecture that integrates gravitational lensing equations into the learning pipeline through a differentiable Singular Isothermal Sphere (SIS) simulator. This system jointly performs classification and lens parameter inference, and serves as the foundation for the proposed project.

In addition, I am currently involved in research at IIT Palakkad on distributed ensemble learning for decentralized datasets, where I study theoretical model risk bounds and build large-scale experimental pipelines in Python.

I have extensive experience with **Python and PyTorch** for building and training deep learning models, and I am comfortable working with modular architectures, experimentation pipelines, and reproducible workflows.

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Code Repository: https://github.com/Abhiraj212024/deeplens_PINN

3 Approaching the Problem

3.1 Problem Overview

Strong gravitational lensing is a powerful probe for detecting dark matter substructures through distortions in observed galaxy images. Although recent machine learning approaches achieve strong classification performance on simulated data, they typically operate as black-box models and do not incorporate the underlying physical processes governing lensing.

This project proposes a **physics-informed learning framework** that integrates gravitational lensing equations directly into the model. By embedding a differentiable physics module within the pipeline, the model is encouraged to learn representations that are both discriminative and physically consistent.

3.2 Model Overview

The proposed architecture consists of three key components: an encoder, a set of prediction heads, and a differentiable lensing simulator.

The framework is **architecture-agnostic**: any suitable encoder (e.g., CNNs or vision transformers) can be used to extract features from the input image. In this work, a ResNet18 backbone is used as a practical baseline due to its stability and efficiency.

The encoder produces a latent representation that is shared across multiple tasks.

3.3 Latent Representation and Prediction Heads

The latent representation is passed through task-specific heads:

- **Classification Head:** Predicts the dark matter substructure class.
- **Physics Head:** Predicts lens parameters, specifically the Einstein radius θ_E and lens center (c_x, c_y) under the SIS model.

- **Decoder:** Reconstructs the source image from the latent representation.

This multi-task design enables the model to jointly learn classification and physically meaningful parameter estimation.

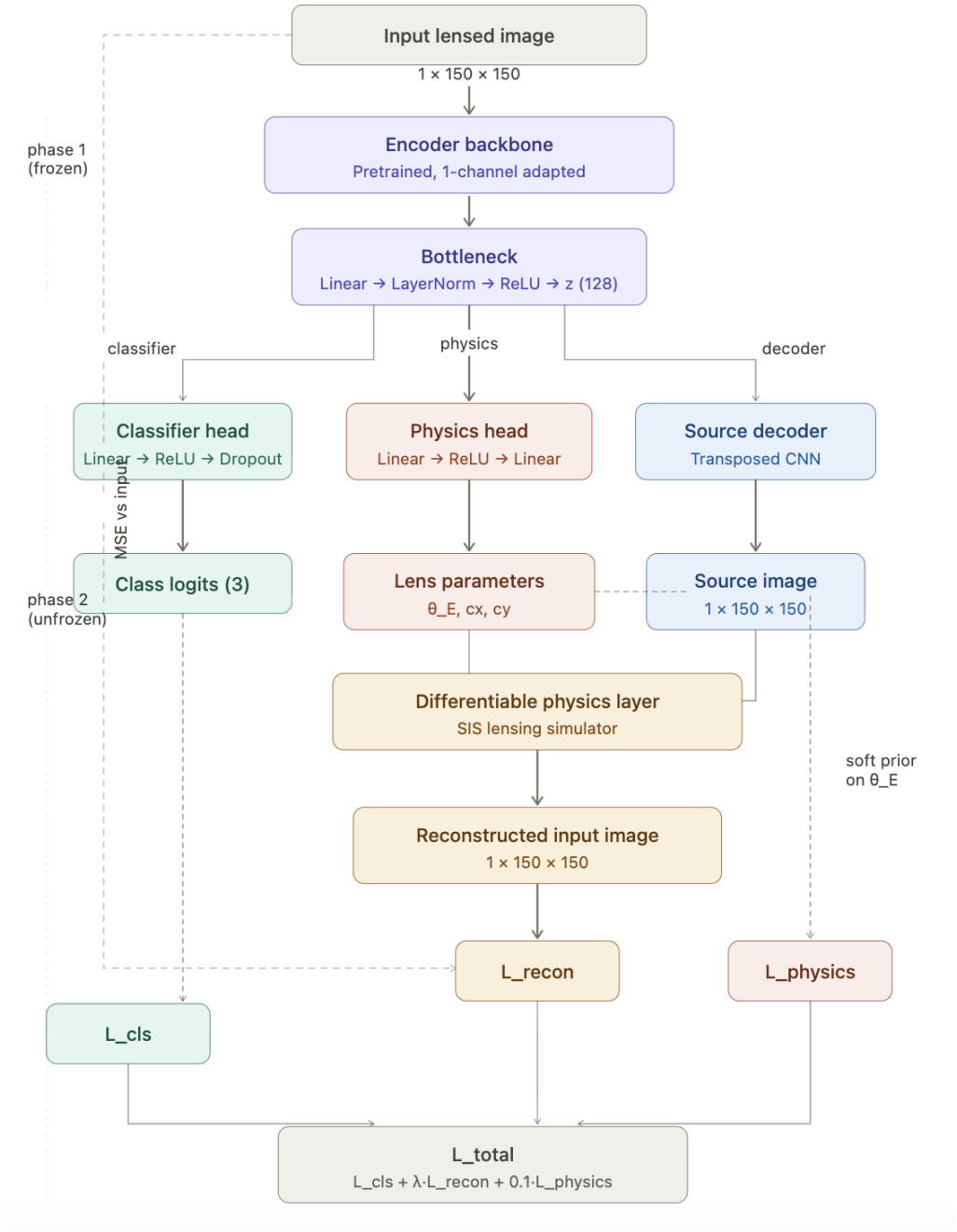

Figure 1: Overview of the physics-informed architecture integrating an encoder with a differentiable lensing simulator.

3.4 Gravitational Lensing Model

The lensing process is modeled using the **Singular Isothermal Sphere (SIS)** approximation. The lens equation is given by:

$$\boldsymbol{\beta} = \boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta}), \quad \boldsymbol{\alpha}(\boldsymbol{\theta}) = \theta_E \frac{\boldsymbol{\theta} - \boldsymbol{c}}{\|\boldsymbol{\theta} - \boldsymbol{c}\|}$$

where θ_E is the Einstein radius and $\boldsymbol{c} = (c_x, c_y)$ is the center of the lens.

These parameters are predicted by the model and used within a differentiable simulator. The framework is **designed to be extensible** to more realistic lens models, such as SIE and NFW.

3.5 Differentiable Lensing Simulation

To enforce physical consistency, the predicted parameters are used to simulate the lensing process:

1. The decoder reconstructs the source image.
2. The predicted parameters (θ_E, c_x, c_y) are applied to the source.
3. A differentiable lensing layer generates a reconstructed lensed image.

This reconstructed image is compared to the observed input. Minimizing this discrepancy encourages the model to learn physically meaningful parameters.

3.6 Training Objective

The model is trained using a multi-task objective:

$$L_{\text{total}} = L_{\text{cls}} + \lambda_{\text{recon}} L_{\text{recon}} + \lambda_{\text{physics}} L_{\text{physics}}$$

where:

- L_{cls}: classification loss (cross-entropy),
- L_{recon}: reconstruction loss (MSE),
- L_{physics}: regularization on predicted physical parameters.

The loss weights λ_{recon} and λ_{physics} control the trade-off between predictive performance and physical consistency.

4 Evaluation Metrics

The proposed model will be evaluated using both machine learning and physics-based metrics to assess classification performance, reconstruction quality, and the estimation of physical parameters. These metrics jointly evaluate predictive performance, reconstruction fidelity, and physical consistency.

Classification Performance: Accuracy, F1-score, and ROC-AUC on validation and test sets across all dark matter substructure classes.

Reconstruction Quality: Mean Squared Error (MSE) between the reconstructed and observed images; Structural Similarity Index (SSIM) for perceptual similarity.

Physical Parameter Estimation: Prediction error for the Einstein radius θ_E and lens center coordinates (c_x, c_y).

5 Preliminary Results

Preliminary experiments were conducted to evaluate the feasibility of the proposed physics-informed architecture. The goal was to determine whether incorporating physical constraints through the lensing equation could preserve classification performance while allowing the model to learn interpretable physical parameters.

5.1 Baseline Comparison

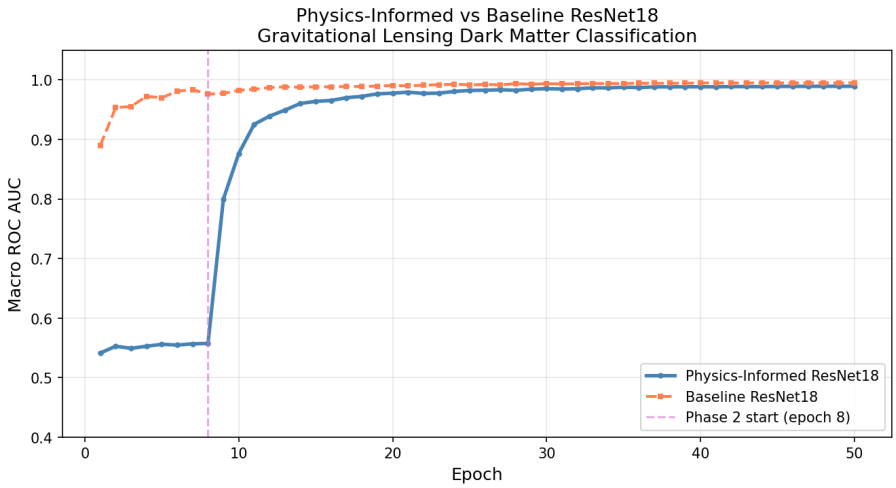

Figure 2: Comparison of classification performance between the baseline ResNet18 model and the proposed physics-informed architecture.

The physics-informed model achieves a macro ROC-AUC of **0.9894** compared to **0.9953** for the baseline ResNet18. This modest gap reflects an expected tradeoff: by constraining the network to produce physically consistent lens parameters, the model sacrifices a small amount of raw discriminative capacity in exchange for interpretable outputs.

These results validate the effectiveness of the physics-informed framework, regardless of the specific encoder choice.

5.2 Reconstruction Quality

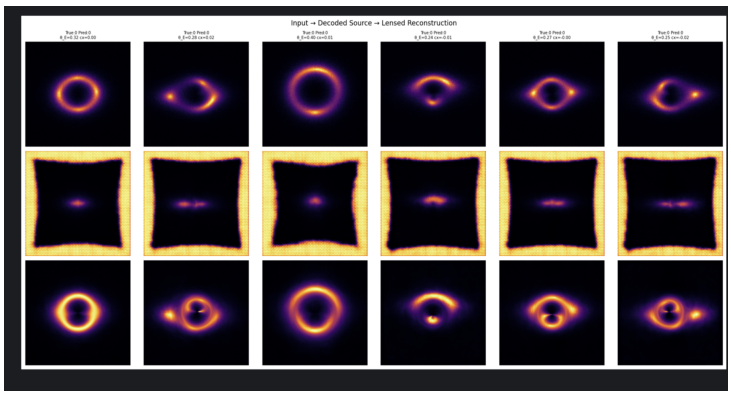

Figure 3: Example reconstructions produced by the physics-informed model. Each triplet shows the input lensed image, the decoded source, and the re-lensed reconstruction.

Reconstructed lensed images generated using the predicted lens parameters closely match the original observations, confirming that the differentiable simulator correctly reflects the governing

lensing equations.

5.3 Physics Parameter Inference

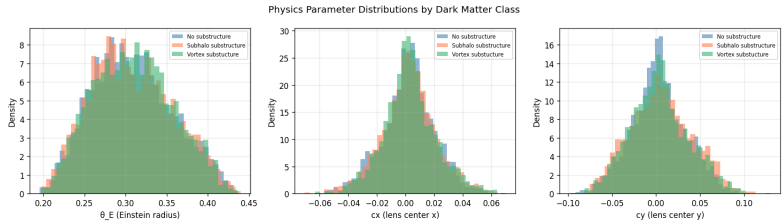

Figure 4: Distribution of predicted lens parameters across dark matter classes. The Einstein radius distributions overlap substantially across classes, as expected for this dataset.

The model predicts the Einstein radius θ_E and lens center coordinates across all three dark matter classes. The predicted θ_E values are consistent with the known data generation parameters (mean ≈ 0.307, std ≈ 0.049), indicating that the physics head has learned physically plausible estimates rather than arbitrary values.

6 Expected Deliverables

- A modular differentiable gravitational lensing layer (SIS, extendable to SIE/NFW) implemented in PyTorch.
- A physics-informed training pipeline that combines classification, reconstruction, and parameter inference.
- Benchmark results comparing physics-informed models with baseline architectures.
- Evaluation pipeline for simulated and observational datasets.
- Well-documented and reproducible code integrated into the DeepLense framework.

7 Proposed Timeline

The project will be carried out over the standard 12-week GSoC development period, with clearly defined milestones and deliverables at each stage.

Community Bonding Period

- Refine the scope of the project and align the architecture design with DeepLense requirements.
- Review relevant literature on gravitational lensing and physics-informed learning.
- Finalize lens modeling approach (SIS with potential extensions to SIE/NFW).
- Identify and prepare datasets (simulated and publicly available observational data such as HST).

Week 1–2: Core Setup and Physics Module

- Implement and validate the differentiable SIS lensing module.
- Set up dataset pipelines, preprocessing, and training infrastructure.

- Define parameter prediction targets (θ_E, c_x, c_y etc.) based on the chosen lensing model.

Deliverable: Working differentiable lensing layer with verified forward pass.

Week 3–4: Architecture Integration

- Integrate the encoder, prediction heads, and decoder into a unified pipeline.
- Connect the physics module with the model outputs.
- Validate end-to-end forward pass and reconstruction pipeline.

Deliverable: End-to-end physics-informed model producing reconstructions and parameter predictions.

Week 5–6: Training and Benchmarking

- Train the multi-task model (classification + reconstruction + physics).
- Tune loss weights and stabilize training.
- Benchmark against baseline models (e.g., ResNet-based classifier).

Deliverable: Trained model with baseline comparison and evaluation metrics.

Midterm Evaluation: Fully functional physics-informed model with validated performance and comparison to baseline.

Week 7–8: Model Extensions and Real Data Evaluation

- Extend the framework to support more complex lens models (SIE/NFW, if feasible).
- Implement preprocessing for observational data (e.g., noise handling, normalization).
- Evaluate model generalization on real or semi-real datasets.

Deliverable: Extended model and initial results on observational data.

Week 9–10: Analysis and Refinement

- Perform ablation studies to evaluate the contribution of physics constraints.
- Analyze trade-offs between classification performance and interpretability.
- Improve model robustness and training stability.

Deliverable: Comprehensive evaluation results and analysis.

Week 11–12: Finalization and Integration

- Refactor and modularize code for integration into DeepLense.
- Prepare reproducible training pipelines and scripts.
- Write documentation, usage guides, and final report.

Deliverable: Clean, documented, and reproducible codebase integrated with DeepLense.

8 Potential Challenges and Mitigation

Training Instability: Joint optimization of classification, reconstruction, and physics losses may lead to unstable gradients or task imbalance. This will be addressed by gradually introducing the reconstruction loss (λ_{recon} scheduling), careful normalization of loss terms, and monitoring task-specific metrics. If instability persists, staged training (pretraining the encoder before enabling the full pipeline) will be used.

Degenerate or Non-Physical Parameter Predictions: The physics head may produce unstable or non-physical parameter values early in training. This will be mitigated using bounded parameterization, regularization based on prior distributions, and, if necessary, auxiliary supervision on lens parameters.

Differentiable Simulator Efficiency: The lensing simulation must remain efficient for end-to-end training. The simulator will be implemented using vectorized GPU operations and optimized tensor computations.

Simulated-to-Real Domain Gap: Models trained on simulated data may not generalize to real observations. This will be addressed through noise modeling, data augmentation, and preprocessing techniques such as normalization and PSF-aware transformations. If needed, fine-tuning on small amounts of real data will be explored.

Model Complexity and Resource Constraints: Multi-task training increases computational requirements. Efficient encoder backbones, lightweight decoders, mixed-precision training, and batch size tuning will be used to ensure feasible training within available resources.

9 References

- Ana Diaz Rivero, Cora Dvorkin, *Direct Detection of Dark Matter Substructure in Strong Lens Images with Convolutional Neural Networks*, arXiv:1910.00015.
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- Michael W. Toomey et al., *LensPINN: Physics Informed Neural Network for Learning Dark Matter Morphology in Lensing*, NeurIPS 2024.
- Michael W. Toomey et al., *HEAL-PINN: Physics-Informed Swin Transformer for Dark Matter Studies for Sparse Lensing Data*, NeurIPS 2025.